

AutoGaze: A Very Initial Exploration in A SAM2-based Pipeline for Automated Eye-Object Interaction Analysis in First-Person Videos

Qing Zhang*
The University of Tokyo

Yifei Huang†
The University of Tokyo

Jun Rekimoto‡
The University of Tokyo
Sony CSL Kyoto

ABSTRACT

This paper presents a novel automated workflow for analyzing eye-tracking data in first-person videos. Our system uses the Segment Anything Model 2 (SAM2) to segment and track objects, correlating them with gaze information to provide a detailed understanding of visual attention and hand-object interactions. The pipeline outputs a structured JSON file containing rich information about gaze behavior, object interactions, and their temporal relationships. We demonstrate the system's effectiveness in a life science operational context and compare its performance with a multimodal Large Language Model (LLM) for automated annotation. Our findings highlight the potential of combining advanced computer vision techniques with LLMs for comprehensive and scalable analysis of gaze behavior in real-world scenarios.

Index Terms: Eye tracking, gaze movement trajectory, automatic event detection, egocentric video processing.

1 INTRODUCTION

Eye tracking has been exhaustively investigated to gain insights into human attention and behavior in a wide variety of fields [5], including human-computer interaction and cognitive psychology. Although state-of-the-art eye tracking devices offer accurate gaze coordinates and fundamental metrics, researchers, particularly in the life sciences, heavily depend on manual analysis to derive significant semantic insights into what objects users focus on and their interactions with these objects over time. This manual annotation process is labor-intensive, susceptible to inconsistencies, and restricts the scale of eye tracking research. To mitigate these challenges, commercial products like Pupil Labs and Tobii offer automated tools and services for primary gaze visualization, such as heatmaps, area of interest analyses, numerical eye movement trajectories, and partly automated event annotations. Nevertheless, neither academic methods nor commercially available services provide consistent linkage of gaze-to-object, trajectories, and interaction events over time, despite their significant potential for uncovering human activity.

Recent developments in computer vision, especially instance segmentation models like Segment Anything Model 2 (SAM2), are paving the way to close this gap by proficiently detecting and following objects across frames. Meanwhile, multimodal LLMs have introduced great potential in both image and video context understanding regarding automatic annotation. For instance, Pupil Labs' Alpha Lab [1] has integrated a multimodal LLM to automate part of the tedious event annotation, though human labelers are still needed to understand the given video context initially.

In this paper, we present an automated workflow that combines eye-tracking data with instance segmentation to provide detailed

*e-mail: qzkiyoshi@gmail.com

†e-mail: hyf015@gmail.com

‡e-mail: rekimoto@acm.org

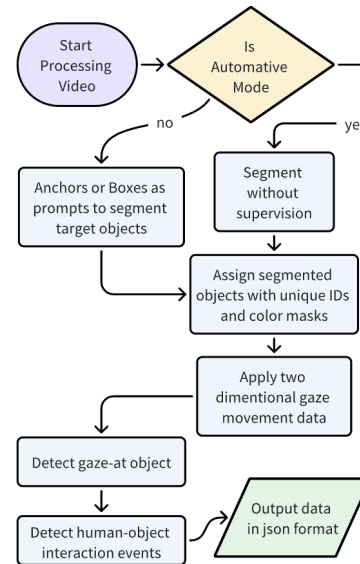


Figure 1: The processing pipeline, illustrating the workflow from initial video input through object segmentation, gaze tracking, and interaction detection, to final JSON output generation. The system supports both automated and prompt-based segmentation modes.

insights into gaze behavior during real-world tasks captured from a first-person perspective. Our system automatically segments and tracks objects in first-person view videos, associates them with gaze information, and monitors interaction events between users and objects. The pipeline outputs a structured format that captures temporal relationships between gaze behaviors and object interactions. We evaluate our system's effectiveness in a life science operational context and compare it with an up-to-date multimodal LLM in automatic annotation.

2 RELATED WORKS

Our research builds on three main areas: automated eye tracking analysis, instance segmentation in egocentric videos, and semantic understanding of human-object interactions.

Automated Eye Tracking Analysis Eye tracking analysis has transitioned from the basic identification of fixations and saccades to more advanced automated methodologies. Notably, Pupil Labs [1] and Tobii [2] have led the way in developing automated tools that create heatmaps and evaluate areas of interest, thereby automating part of the process in conventional manual analysis. Recent developments by Panetta et al. [9] introduced Gaze to Object Classification (GoC) software that eliminates the necessity for manual annotation in eye-tracking data analysis, offering precise and affordable outcomes. Similarly, Negar Alinaghi et al. [3] presented

a method that combines YOLOv8 and Mask2Former to automatically annotate fixation points in eye-tracking videos.

Video Annotation Using LLM Recent research has shown significant advancements in leveraging Large Language Models (LLMs) for automating video annotation tasks. The HowToCaption framework demonstrates how LLMs can transform ASR subtitles into high-quality video descriptions, enabling large-scale annotation without human supervision [13]. Similarly, LAVILA introduces an approach where LLMs are repurposed as visually-conditioned narrators, generating dense coverage of long videos with better temporal synchronization [17]. Studies have shown that LLMs exhibit higher accuracy in object and activity annotations compared to emotion and genre annotations when analyzing video content [7]. The Video-ChatGPT system employs a hybrid approach combining human-assisted and semi-automatic annotation methods to generate comprehensive video instruction data [8]. These approaches have proven particularly effective for instructional videos, where traditional ASR-based annotations often provide noisy supervision due to the natural divergence between speech and visual content [13]. The automation of video annotation through LLMs not only reduces the costs associated with manual labeling but also enables the creation of larger, more diverse datasets that can improve the performance of text-video retrieval and video captioning tasks [6, 14].

Instance Segmentation and Understand Interaction in Egocentric Videos The development of advanced instance segmentation models has greatly enhanced our capacity to analyze egocentric videos. Meta's Segment Anything Model 2 (SAM2) [10] marks a substantial progression in this area, providing strong object segmentation capabilities with minimal initialization needs. In addition, approaches based on YOLO series have demonstrated potential for real-time object detection and classification within egocentric videos [11, 4]. For detecting and understanding interactions between humans and target objects, Shan et al. [12] introduce a method that can deduce a comprehensive depiction of hands involved in interaction, including hand location, orientation, contact status, and an enclosing box around the contact object, using internet videos.

3 APPROACH DESIGN

Our method analyzes first-person view (FPV) video aligned with eye-tracking information using a multi-step process to achieve a semantic interpretation of gaze patterns. The system consists of four primary modules: (1) video preprocessing and object initialization, (2) ongoing object tracking and segmentation, (3) correlating gaze with objects, and (4) analyzing semantic interactions. The anticipated output format is outlined in Table 1.

The preprocessing phase starts with an initialization step where SAM2 develops a semantic comprehension of its surroundings. Our system supports both the automatic prompting feature and a prompt-based interface in SAM2 to effectively balance the efficiency and precision of object segmentation. An example of labeling targeted objects can be seen in Figure 2. The detailed workflow is depicted in Figure 1. During initialization, unique identifiers are assigned to each detected object, accompanied by a confidence score derived from SAM2's segmentation quality metrics. These identifiers are retained throughout the session, permitting the temporal tracking of objects even when they momentarily exit the field of view. Post-initialization, SAM2's per-session memory module ensures object consistency across frames. To manage situations where objects become occluded or exit the field of view for extended durations, we employ a system that keeps an ongoing log of tracking confidence scores and any segmentation failures, aiding researchers in pinpointing potential uncertainties in the analysis. Regarding gaze-object correlation, eye-tracking data, including raw gaze coordinates and derived metrics (fixations, saccades),

Table 1: Desired Output Data Structure Description

Field	Description	Example Value
session_id	Unique identifier for each recording session	task_1_p03
timestamp	Unix formatted time of the gaze event	1736939069
coordinates	[x, y] pixel coordinates of gaze position	[320, 240]
target_object_id	Unique identifier of the gazed object	pipette_01
confidence_score	SAM2's confidence in object detection	0.92
interaction_type	Classified type of gaze interaction	Hand_index interacting with Object_index
duration	Duration of the gaze event in seconds	1.5
behavior_pattern	High-level interpretation of behavior	pipette operation
related_objects	List of objects involved in the interaction	["culture dish_01", "pipette_01"]

are synchronized with the segmented video frames. For each gaze point, the system: (a) Maps the gaze coordinates to the corresponding location within the segmented frame, (b) Identifies the object (if present) at that location using the unique identifiers, (c) Records temporal information such as timestamp, duration, and transition patterns. This process is expected to generate a continuous series of gaze-object interactions, forming the foundation for higher-level semantic analysis. The concluding stage might utilize EgoHOS [16] to decode the sequence of gaze-object interactions. This component: (a) Analyzes the temporal sequence of gaze-object correlations, (b) Recognizes significant interaction patterns (e.g., sequential scanning, focused attention), and (c) Produces semantic annotations describing the observed behaviors. The expected output is formatted as a JSON file containing multiple keys and values, as detailed in Table 1.

4 INITIAL IMPLEMENTATION

In our initial implementation of the proposed method, we utilized a recorded first-person-view (FPV) video depicting a typical life science operation in conjunction with corresponding two-dimensional eye-tracking data. The gaze data was downsampled to 30 Hz to match the FPV video. This recording was captured using the Pupilabs NEON eye tracker and lasted approximately two minutes and three seconds. The operational environment was entirely dependent on artificial lighting. Upon analyzing the results, our method generally succeeded in maintaining the tracking of manually segmented objects across the frames, as illustrated in Figure 3. However, there were instances of losing track of uniquely assigned objects, primarily owing to the inherent limitations [15] of SAM2, which is only capable of handling short-term occlusions of targets. The inability to distinguish between the left and right hand indicates the necessity for users to deliberately position their hands in a manner conducive to SAM2 (clearly and separately placed) along with considering the arrangement of other target objects. With respect to the detection of interaction types, our modified approach benefits from



Figure 2: An example of labeling targeted objects.

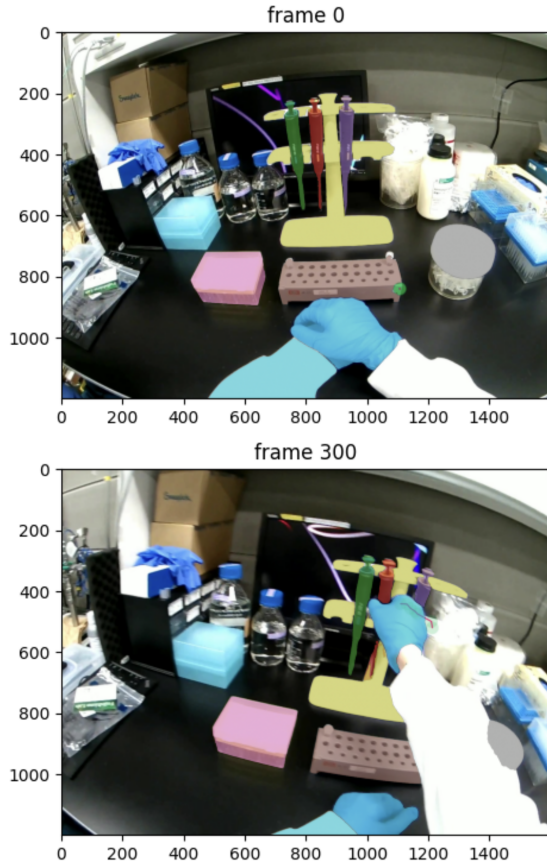


Figure 3: An example of tracking targeted objects over 300 frames.

the stable performance of EgoHOS [16] in identifying hand-object interactions. Nevertheless, relying on hard-coded events may still be either time-intensive or insufficiently detailed to accurately represent real-world hand-object manipulation and operation. Object recognition of the equipments in the scene of life-science operation also falsely labeled the detected objects due to its hard-coded-label nature. Additionally, inadequate dynamic range and poor white balance control further degraded the system's performance in tracking objects. For example, the initially clear boundaries of transparent containers appeared as a cluster of white fragments. Furthermore, the built-in white balance control of the eye tracker caused the colors of objects to change significantly, leading to noticeably different colors for the same objects across frames.

4.1 Large Language Model based Annotation in Comparison

Incorporating the challenges encountered in our earlier examination of the proposed approach, we consistently utilized the newly launched Gemini 2.0 Flash Experimental, exclusively via Google AI-studio, to annotate the same FPV video. This ensured that the output was aligned with our proposed method's format. Below, we provide two excerpts from the resulting JSON:

```
{
  "session_id": "task_1_p03",
  "timestamp": 0.6,
  "coordinates": [ 450, 450 ],
  "target_object_id": "pipette_01",
  "confidence_score": 0.92,
  "interaction_type": "observed",
  "duration": 1.2,
  "behavior_pattern": "adjusting pipette",
  "related_objects": []
},
{
  "session_id": "task_1_p03",
  "timestamp": 6.1,
  "coordinates": [ 450, 250 ],
  "target_object_id": "pipette_01",
  "confidence_score": 0.92,
  "interaction_type": "observed",
  "duration": 1,
  "behavior_pattern": "adjusting pipette",
  "related_objects": []
},
}
```

While the "timestamp," "duration," "(gaze) coordinates," and "confidence_score" were imprecisely measured and not directly relevant in this LLM-based annotation, the "target_object_id" and "behavior_pattern" exhibited enhanced accuracy and adaptability in analyzing and explaining the current task when compared to our method relying on predefined labels. The inaccuracy in timestamp and gaze duration is probably due to differences in frame sampling ratios and analytical techniques between our approach and the LLM utilized.

5 DISCUSSION

In this initial exploration of our proposed approach and a multi-modal Large Language Model (LLM) for automating annotation, we observed several key insights and challenges. Our approach, which leverages SAM2 for object segmentation and tracking in first-person-view (FPV) videos, demonstrated relatively good performance in maintaining the tracking of manually segmented objects across frames. However, several limitations were identified, particularly in scenarios involving changes in white balance, frequent object occlusion, and crowded environments with similar objects. These challenges highlight the need for further refinement in

object tracking and segmentation, especially in dynamic and complex scenes. This limitation is particularly evident in life science operational contexts, where objects such as transparent containers or similar-looking equipment can lead to tracking failures. Future work could explore the integration of additional computer vision techniques, such as Kalman filters [15], to improve object tracking in highly dynamic scenes. Additionally, the system's inability to distinguish between left and right hands necessitates careful hand positioning by users, which may not always be feasible in real-world scenarios.

In comparison, the LLM-based annotation approach, specifically using Gemini 2.0 Flash Experimental, showed enhanced accuracy and adaptability in identifying target objects and interpreting behavior patterns. The LLM's ability to generate more nuanced and contextually relevant annotations suggests that integrating LLMs into our pipeline could significantly improve the semantic understanding of gaze-object interactions. However, the LLM-based approach also faced challenges, particularly in accurately measuring timestamps and gaze durations, which are crucial for temporal analysis. This discrepancy is likely due to differences in frame sampling ratios and analytical techniques between our approach and the LLM.

The preliminary results suggest that a hybrid approach, combining the strengths of SAM2 for object tracking as well as other quantitative data and LLMs for semantic annotation, could offer a more robust solution for automating gaze behavior analysis. For instance, SAM2 could handle the initial object segmentation and tracking, while LLMs could provide higher-level semantic interpretations of gaze-object interactions. This combination could mitigate the limitations of each method when used in isolation, leading to more accurate and comprehensive annotations.

In conclusion, while our proposed approach shows the potential in automating gaze behavior analysis, there is significant room for improvement, particularly in handling complex and dynamic environments. The integration of LLMs and advanced computer vision techniques could provide a more robust and scalable solution, enabling more extensive and uniform analysis of gaze behavior in real-world tasks. Future research should focus on addressing the identified limitations and exploring hybrid approaches that combine the strengths of different methodologies to achieve more accurate and comprehensive results.

ACKNOWLEDGMENTS

This work was partially supported by JST Moonshot R&D Grant JPMJMS2012, and a project(JPNP23025)commissioned by the New Energy and Industrial Technology Development Organization (NEDO).

REFERENCES

- [1] Alpha Lab - Home - Pupil Labs Docs — docs.pupil-labs.com. <https://docs.pupil-labs.com/alpha-lab/>. [Accessed 03-01-2025]. 1
- [2] Global leader in eye tracking for over 20 years — tobii.com. <https://www.tobii.com/>. [Accessed 03-01-2025]. 1
- [3] N. Alinaghi, S. Hollendonner, and I. Giannopoulos. Myfix: Automated fixation annotation of eye-tracking videos. *Sensors*, 24(9):2666, 2024. 1
- [4] M. Hussain. Yolo-v1 to yolo-v8, the rise of yolo and its complementary nature toward digital manufacturing and industrial defect detection. *Machines*, 11(7):677, 2023. 2
- [5] P. Kiefer, I. Giannopoulos, M. Raubal, and A. Duchowski. Eye tracking for spatial research: Cognition, computation, challenges. *Spatial Cognition & Computation*, 17(1-2):1–19, 2017. 1
- [6] N. Kudan. Video annotation: challenges and best practices — toloka.ai. <https://toloka.ai/blog/video-annotation/>, 2023. [Accessed 28-01-2025]. 2
- [7] J. Liu, Y. Wang, Y. Lyu, Y. Su, S. Niu, X. O. Xu, and Y. Zhang. Harnessing llms for automated video content analysis: An exploratory workflow of short videos on depression. In *Companion Publication of*

the 2024 Conference on Computer-Supported Cooperative Work and Social Computing, pp. 190–196, 2024. 2

- [8] M. Maaz, H. Rasheed, S. Khan, and F. S. Khan. Video-chatgpt: Towards detailed video understanding via large vision and language models. *arXiv preprint arXiv:2306.05424*, 2023. 2
- [9] K. Panetta, Q. Wan, A. Kaszowska, H. Taylor, and S. Agaian. Software architecture for automating cognitive science eye-tracking data analysis and object annotation. *IEEE Transactions on Human-Machine Systems*, 49:268–277, 2019. doi: 10.1109/THMS.2019.2892919 1
- [10] N. Ravi, V. Gabeur, Y.-T. Hu, R. Hu, C. Ryali, T. Ma, H. Khedr, R. Rädle, C. Rolland, L. Gustafson, E. Mintun, J. Pan, K. V. Alwala, N. Carion, C.-Y. Wu, R. Girshick, P. Dollár, and C. Feichtenhofer. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024. 2
- [11] J. Redmon and A. Farhadi. Yolo3: An incremental improvement. *arXiv e-prints*, pp. arXiv–1804, 2018. 2
- [12] D. Shan, J. Geng, M. Shu, and D. F. Fouhey. Understanding human hands in contact at internet scale. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 9869–9878, 2020. 2
- [13] N. Shvetsova, A. Kukleva, X. Hong, C. Rupprecht, B. Schiele, and H. Kuehne. Howtocation: Prompting llms to transform video annotations at scale. In *European Conference on Computer Vision*, pp. 1–18. Springer, 2025. 2
- [14] X. Wu, S. Huang, G. Wang, J. Xiong, and F. Wei. Boosting text-to-video generative model with mllms feedback. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024. 2
- [15] C.-Y. Yang, H.-W. Huang, W. Chai, Z. Jiang, and J.-N. Hwang. Samurai: Adapting segment anything model for zero-shot visual tracking with motion-aware memory, 2024. 2, 4
- [16] L. Zhang, S. Zhou, S. Stent, and J. Shi. Fine-grained egocentric hand-object segmentation: Dataset, model, and applications. In *European Conference on Computer Vision*, pp. 127–145. Springer, 2022. 2, 3
- [17] Y. Zhao, I. Misra, P. Krähenbühl, and R. Girdhar. Learning video representations from large language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 6586–6597, 2023. 2